

# A Review of Silicon and Organic solar cells: Structures and Efficiency Comparisons.

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**Abstract**-This review explores the structural designs and Efficiency metrics of silicon solar cells including monocrystalline, polycrystalline and Amorphous types, and also Organic silicon solar cells. They have recently gained interest due to its light weight properties, flexibility and potential low-cost production. this paper provides analysis of the structures and their efficiencies of both silicon and Organic solar cells.

## I. INTRODUCTION

In today's world solar energy play a vital role in combating climate change and achieving global energy goals. Solar energy is the topmost renewable energy resource which is widely used across world. Increased energy consumption and depletion of fossil fuels resources, beside global climate change warnings, lead to increasing concern in renewable energy.

So, to minimize the effects of climate change the world now investing into renewable energy sources, the most available renewable energy resource today is solar energy where it is available around the world, for gaining maximum amount of electricity from the solar energy, it all depends upon the solar cells and their structures This paper mainly describes about the silicon and Organic solar cells, their structures and efficiencies. There is chart[fig.1] of the highest confirmed conversion efficiencies for research cells for a range of photovoltaic technologies, plotted from 1976 to the present.

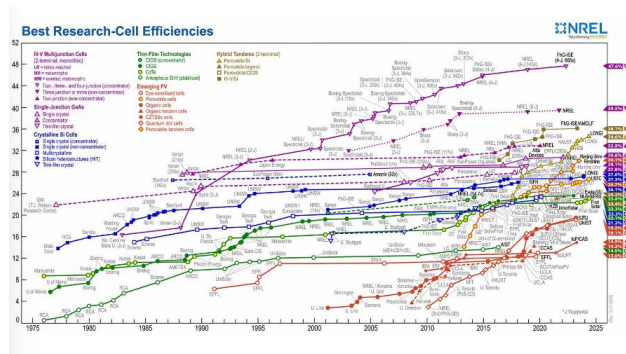


Fig.1. Best-Research cell efficiencies.

## II. a. Mono Crystalline Solar Cells

Mono Crystalline solar cells are constructed from single crystals usually in the form of ingots sliced into a number of cells. The entire ingot is doped, usually P-type with boron

during melt phase and sawn into circular wafers with the thickness less than 0.5mm. An N-type dopant, usually phosphorus is then diffused into the surface to form the P-N junction, as shown in fig.2. The process of production these crystals is known as Czochralski, first silicon is in the form of Trichlorosilane, which contains large number of unwanted elements, these elements are removed in further processes [1,2,3].

Heavily doped layers are formed at each surface to aid electrical contact and the metal contacts are formed by the screen printing of a metallic paste before firing. Optical coatings may be added to the front surface of the wafer [3]. These solar cells have numerous benefits, such as better efficiency than other sorts of solar cells, cell efficiency of mono crystalline solar cells is 15% to 20%. Disadvantage of these solar cells, is complicated manufacturing process is required to produce mono crystalline silicon, Reduction in efficiency as temperature drops to approximately 25°C [1,2,3].

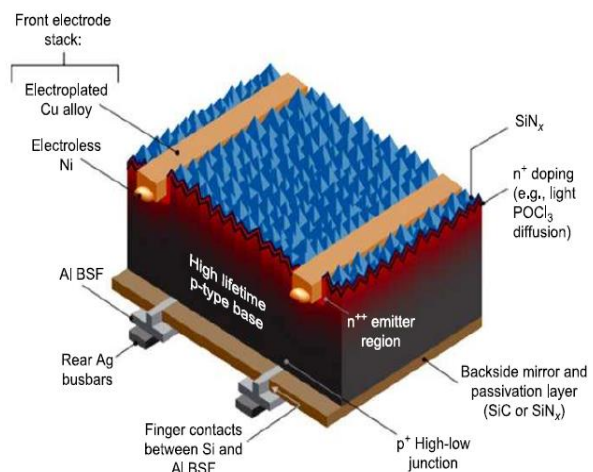


Fig.2. schematic of mono crystalline PV cell [3].

## b. Poly Crystalline Silicon Solar Cell

Poly crystalline silicon solar cell is a high purity poly-Si form. A chemical purification process known as siemens process produces poly-Si from metallurgical grade silicon, these materials are produced specifically for PV production and have advantage that the square wafers make cells that pack together closely in modules. Leaving less unproductive space between them, Poly crystalline solar cells are less expensive than mono crystalline solar cells but also less efficient, because of their

non-uniform lattices, they have number of ingots drawn from molten vat rather than large ingot as shown in Fig.3. Poly crystalline materials are in less efficient due to pure electronic qualities of the boundaries between the randomly oriented silicon grains (crystals) [4]. These cells have grain boundaries (A grain boundary is a two-dimensional defect in a poly crystalline material that occurs at the interface between two grains (crystals) with different crystallographic orientations) as shown in Fig.4. because of these grain boundaries an additional source of recombination takes place, so it is expected that the efficiency of these cells is lower than that of their mono crystalline counterparts [3,12]. An additional disadvantage for reduced performance is difficult in texturing such material, and commonly multi crystalline cells are untextured. (Texturing in solar cells is a process that involves etching the surface of silicon wafers to create pyramids which reduces light reflection and improves cell efficiency), texturing reduces light reflection, which allows photons to be absorbed by the solar cell, this increases cells short circuit current density. The efficiency of typical poly solar cell is 8 to 12%.

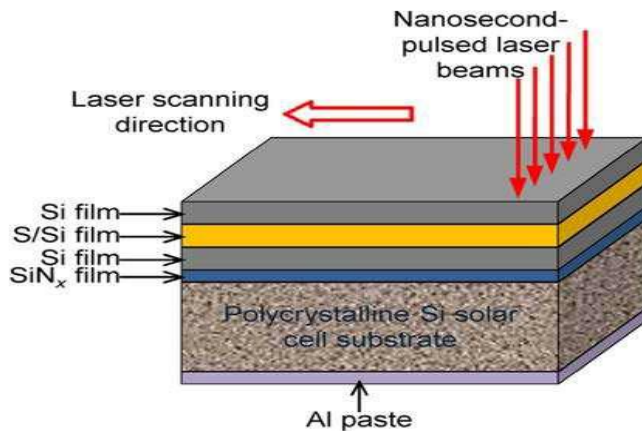


Fig.3. Structure of typical Poly-Si solar cell [3].

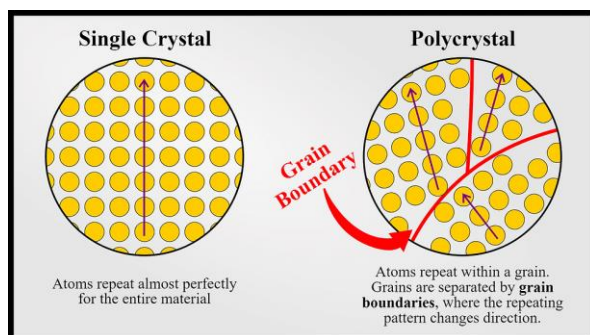


Fig.4. Grain boundary defect in Poly – Si solar cell.

### C. Amorphous Silicon Solar Cells

Amorphous silicon solar cells are type of thin-film solar cells made using non crystalline or Amorphous silicon, which is different from crystalline silicon (C-Si). Amorphous silicon atoms (a-Si) arranged more randomly, where in the crystalline silicon atoms we can see strict geometric structures or orders, Many atoms have bonds that are unsatisfied by neighboring silicon atoms, so, hydrogen is usually incorporated into the

material to reduce recombination, (a-Si-H). The material now which is called as hydrogenated amorphous silicon as shown in Fig.5 [1,2,3,11].

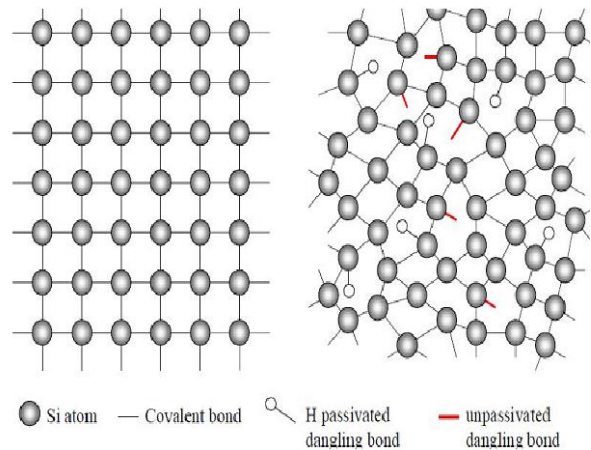


Fig.5. Atomic model structure of Crystalline silicon and hydrogenated amorphous silicon [4].

It has high density of defects that would prevent the material from exhibiting minimal transport and recombination properties, but due to hydrogen incorporated into those silicon bonds that are free and which are mainly responsible for the poor transport and recombination properties. Compare to crystalline material amorphous silicon is much more absorbent. a-Si-H is usually applied by plasma enhanced chemical vapor deposition, it has P-i-N structure, with a layer of intrinsic material between the p and N typed layers as shown in Fig .6 [4].

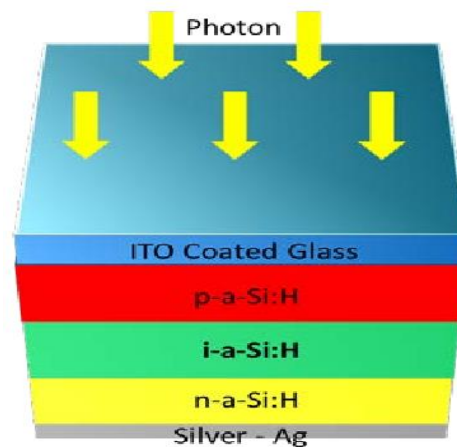


Fig.6. Schematic structure of a-Si-H (p-i-n) junction solar cell.

Here p-i-n junction design, the electric field extends across the intrinsic region, permitting more thickness for generation and collection within field region, Major disadvantage is degradation, the process of degradation of these cells is known as Staebler-Wronski effect, it is minimized by making the cells as thin as possible and textured TCO surfaces has been used to recover the absorptivity lost by thinning, where Ge is alloyed with Si to modify bandgap of around 1.7ev is higher than optimal, advantage of Amorphous silicon solar cells is that by alloying it with nitrogen or carbon the value of its band gap can

be increased. Amorphous modules have small negative temperature coefficient of power (-0.1%). So because of this feature these cells are manufactured by using low temperature deposition technology allows plastics to be used as substrates in flexible modules, therefore the efficiency of these cells are limited between 6% to 8% [3,4,11].

#### D. Organic solar cells

Organic Solar Cells are 3-dimensional solids made of polymers (or) small organic molecules mainly consisting of carbon and hydrogen with a few additional elements such as sulfur and nitrogen, these materials are not crystalline i.e., they do not form a periodic lattice, neighboring molecules interact weakly with one another so, electrons of each molecule remain confined to it, rather than shared with the surroundings. In inorganic semiconductors the energy band diagram has narrower energy widths as shown in Fig.7. here only photons with an energy close to the band gap is absorbed [5,16].

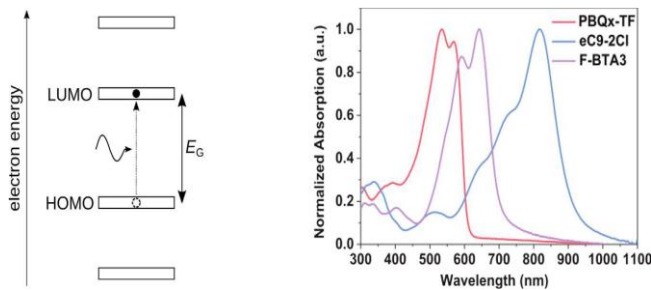


Fig.7. (left) energy band of OSC, (right) Absorption spectra of OSC [5].

The working principle of OSC's are not similar to conventional p-n junction, instead a combination of two different OS are used, one with higher HOMO (highest occupied molecular orbital) and LUMO (lowest unoccupied molecular orbital), called donor and with lower HOMO and LUMO called acceptor. Here a photon is absorbed in an OS, the generated electron and hole remain strongly attached to each other. Such a tightly bound electron-hole pair cannot produce photocurrent they cannot remain tightly bound as shown in Fig.8., electrons and holes should move independently and reach opposite contacts. To split these pairs, a donor-acceptor heterojunction is used. OSC's have high absorption coefficients, making it possible to fabricate absorber layers for solar cells as thin as 100nm [5,15].

In comparison to Si-based PV systems, OSC's have emerged as the least expensive ones in which carbon-based semiconducting material is used as an active layer.

Fig.8. (left) Band diagram of a donor-acceptor in heterojunction OSC's, (right) Typical layer structure of OSC's.

The most commonly used architectures of OSC's are an active layer is sandwiched between ETL (electron transport layer) and HTL (holes transport layer) as shown in Fig.9. a conventional OSC is composed of 6-layers, layer-1 is cathode, layer-2 is ETL, layer-3 is active layer, layer-5 is HTL and layer-6 is transparent anode, last layer is a substrate. Transitional metal oxides such as ZnO, TiO<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub> are used in layer-2,

indiumtinoxide(ITO)/poly(3,4ethylenedioxythiophene):(styrene sulfonate) is used as active organic layer, MoO<sub>3</sub> is used in Transparent anode layer[5].

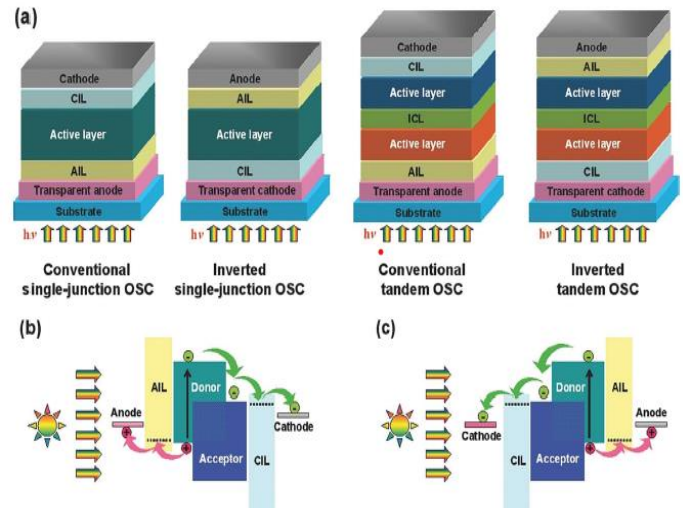


Fig.9. Conventional structures of OSC's [5].

The transportation of electrons and holes play a important role in the output power efficiency of OSC's, due to the large band gap, metal oxides acts as barriers for holes or electrons for providing extremely selective contacts for extracting other types of carriers. The usage of metal oxides in OSC's improves the efficiency [14,16].

Efficiency of solar cell is calculated by:

$$\eta = (\text{Voc} \times \text{Isc} \times \text{FF} / \text{G} \cdot \text{A}) \times 100$$

where:

- Voc is the open-circuit voltage,
- Isc is the short-circuit current,
- F is the fill factor,
- G is the solar irradiance (typically 1000 W/m<sup>2</sup> under standard test conditions),
- A is the area of the solar cell.

Due to disadvantage of tightly bound electron-hole pair in the conventional OSC's efficiency of the cells decreases to overcome this an additional dissociation energy is needed in OSC's. Therefore, a second layer of organic material is required to add with original organic layer for fast dissociation of these electron-hole pair. Illustration of active organic layers are shown in fig.10.

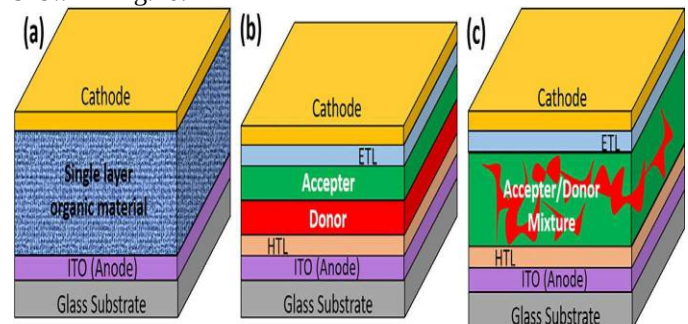


Fig.10.(a) single layer structure (b) Acceptor and donor planar structure (c) Heterojunction structure [5].



An OSC's is a variety of the PV solar cell, one of the OSC's organic layer has a chemical structure as shown in Fig.11. Performance of this cells increased when we added the additives as shown in table .1.

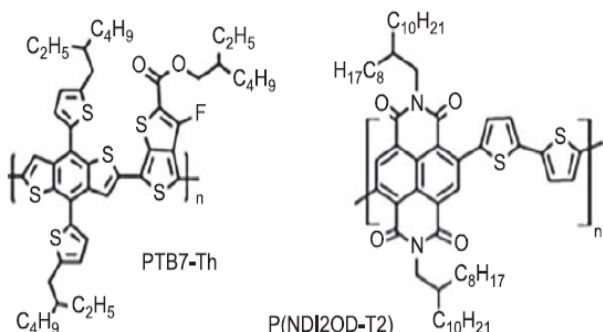


Fig.11. Chemical structure of active organic layer [5].

| Condition     | $V_{oc}$ (V) | $J_{sc}$ ( $\text{mAcm}^{-2}$ ) | Fill factor (FF) | $\eta$ (%) |
|---------------|--------------|---------------------------------|------------------|------------|
| W/o additive  | 0.80         | 11.92                           | 0.47             | 4.51       |
| With additive | 0.80         | 12.64                           | 0.55             | 5.41       |

TABLE.1. Performance parameters of Organic solar cells with and without additives.

The inverted organic solar cells using hafnium indium-zinc-oxide (HIZO) act as an electron transport layer with active layer of bulk heterojunction of poly[4,8-bis(2-ethylhexyl) carbonyl] thieno [3,4-b] thiophenediyl and (PC70BM) and PFN as ETL as shown in Fig.12. The degradation of cell with HIZO-OSC is slow as compared to the cell with PFN [3,7,17].

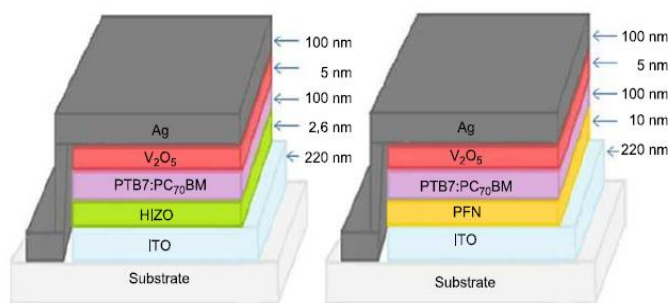


Fig.12. Schematic structure of iOSC's HIZO as ETL and PFN as ETL[5].

The organic cells based on PTB7Th:PC71BM utilized thiophene (TH) and diphenyl ether (DPE) as halogen-free processing solvents. Power conversion efficiency of 9.02% was achieved by thiophene (TH) where as only 7.41% was obtained from chlorobenzene (CB) as mentioned in table.2. The cost effectiveness and flexibility are the main advantages of these solar cells.

| Solvent | $V_{oc}$ (V) | $J_{sc}$ ( $\text{mAcm}^{-2}$ ) | Fill factor (FF) | $\eta$ (%) |
|---------|--------------|---------------------------------|------------------|------------|
| CB      | 0.76         | 15.40                           | 0.62             | 7.41       |
| TH      | 0.81         | 16.60                           | 0.66             | 9.02       |

TABLE.2. Photovoltaic parameters of OSCs using chlorobenzene (CB) and thiophene (TH).

The typical efficiency of Organic Solar Cells lies between 8% to 10 % [5,17].

## II DISCUSSION

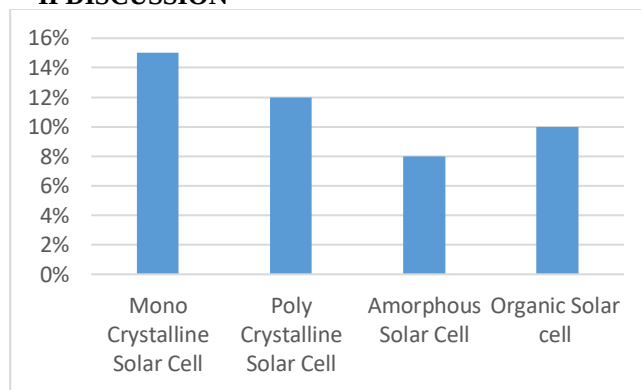


Fig.13. Bar chart comparing the efficiencies of solar cells reviewed in this paper.

SUMMARY-This paper discusses various structures and their efficiencies of solar cells.

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